RevTrace: Preserving Valid Progress across Trusted Feedback Revisions

Anonymous submission
Regular paper

***Abstract*—Trusted feedback can invalidate speculation without invalidating all of its computation or delivered progress. RevTrace composes dependency reuse and whole-hypothesis consumption around a protected feedback-state contract. Complete frozen inputs provide executable eligibility witnesses; compatible certified updates preserve supported ahead views while original selection retains write authority. A native implementation confirms inputs of already executing full recognizer–corrector calls without requiring their own next predictions to be correct. In a predeclared 24-speaker ASR stratum, computation-only retention increases frontier withdrawals from 12 to 17; joint coordination produces zero and adds 26.645 seconds of deadline credit over 714.110 seconds offered. On four previously unused native speakers, full-call retention meets both whole-run deadlines in eight of eight runs versus zero for conservative execution, preserving complete original correction histories. A 48-run workload study separates timely first delivery from subsequent availability and identifies measured service boundaries.**

***Index Terms*—dependable inference, feedback state, incremental computation, speculative execution, streaming ASR**

I. Introduction

Feedback-driven speech recognition repeatedly turns generated text into input for later inference. A trusted correction can improve the current transcript and change the state from which the next recognition starts. This makes each correction an execution boundary: some ongoing speculative work has become invalid, while other work may still be a valid continuation. A dependable service must distinguish these cases while delivering useful progress before its trusted path catches up.

An execution from our prototype exposes the problem. A complete hypothesis covering a two-second audio interval reached the actual Consumer 1.444 s after that interval became available, meeting its two-second display deadline. At 1.544 s, a new certified result replaced the visible view with a shorter snapshot. The discarded hypothesis still had a valid dependency path from the new feedback state. It remained absent at the original two-second deadline and returned at 3.720 s through certified delivery, within its eight-second deadline and with exactly the same text. The temporary loss removed 37 newly displayed characters, including spaces. Both first-delivery deadlines were met, yet already delivered valid progress disappeared for 2.176 s. This is a loss of interval coverage under a compatible update, independent of whether the provisional transcription is semantically correct.

Computation retention alone can exacerbate this mismatch. In our predeclared Qwen evaluation on three new eight-speaker cohorts, retaining valid computation improved timely delivery but increased observed audio-frontier withdrawals from 12 to 17 when the Consumer still installed snapshots. Retaining the computation and its compatible whole view together produced zero frontier withdrawals in that stratum. The issue is therefore how valid work becomes sustained service across a trusted update, as well as how quickly the work finishes.

There are strong foundations for this design. Incremental dialogue models already distinguish active and committed information and propagate revocation [1]; incremental computation reuses work through dependencies and stable identities [2]–[4]. SSBD coordinates old draft reuse with display-only masking in re-translation [5], and PredGen speculates on downstream responses while speech input evolves [6]. We apply these ideas to a protected feedback contract in which the original trusted selection recursively supplies later ASR inputs. The contract governs two decisions: whether executed work remains eligible under the new root, and whether installing that root preserves the Consumer's current whole hypothesis. Separate sufficient guards enforce the feedback dependencies at these boundaries. Together they provide conditional continuity of already delivered progress while retaining selection authority and necessary revisions.

RevTrace reconciles executed calls and delivered whole hypotheses with each new trusted feedback state. A call remains eligible when its complete frozen input has a valid dependency witness from that state; the Consumer separately checks whether its current whole hypothesis remains supported. These decisions retain work and service while original protected selection determines the next feedback state. We instantiate the same eligibility principle with recorded continuation paths in the core runtime and with input-confirmed full calls in a native correction pipeline. Work with no remaining use follows a separate coverage rule, and its physical lease ends only after actual drain.

This paper makes three contributions:

- A feedback-state contract and executable sufficient witnesses for retaining owned computation across trusted updates, including recorded continuations and input-confirmed full calls. These witnesses bind complete state, actual audio frontiers, frozen inputs, and execution identity.
- Joint coordination of computation and whole Consumer views, with a conditional guarantee that applying a compatible certified root does not itself withdraw an ahead view. Necessary revisions remain permitted.

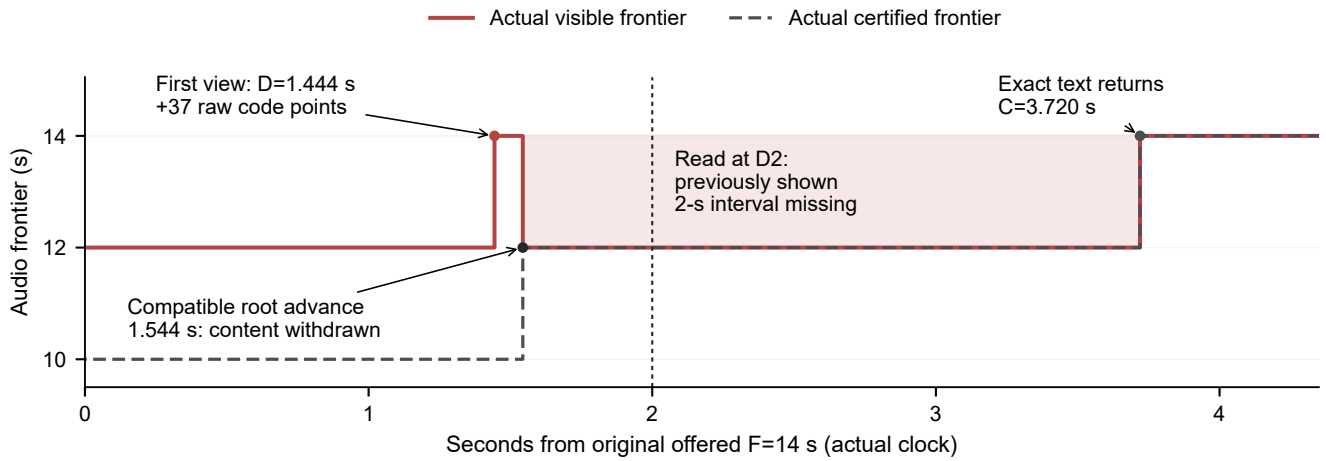


Fig. 1. An actual interval in the computation-only arm. First display meets D2 and certified delivery meets C8, yet a compatible root update withdraws the newly displayed phrase for 2.176 s. The dotted line samples the actual view at the original D2 deadline. This explanatory case comes from the c16 workload, whose whole-run D2 P99 fails; it is not a paired counterfactual or an estimate of natural prevalence.

- An implementation and service evaluation that hold the selected feedback task fixed. Factorial controls isolate computation and whole-view retention; a published native correction pipeline tests whether input-confirmed full execution improves timely service while preserving original correction and complete state transitions.

In the independent headline stratum, the joint design adds 26.645 s of original dual-deadline credit over 714.110 s offered audio while preserving the complete certified histories. A separate 48-run workload study shows where the arms converge and where deadline pressure remains, including a negative intermediate-load confirmation. Subsequent-availability analysis keeps its own definition and leaves the original first-delivery metrics unchanged. A separately frozen native confirmation on four previously unused speakers raises credit from 37.0 to all 66.340 s offered per arm and whole-run D2/C8 passes from zero to eight of eight runs. The core study isolates whole-view continuity; the native study tests timely execution of the unchanged original feedback task.

II. Feedback Contract

A. Feedback creates an execution dependency

Consider a service that receives an immutable audio stream and repeatedly uses generated output as input to later recognition. A protected selection path produces *Canonical* facts. A faster *Private* path may compute and display provisional hypotheses while selection is pending. The selected correction can change the state consumed by the next call: StreamCorrect, for example, feeds its chosen correction back into subsequent ASR [7]. The problem is to coordinate this dependency while the two paths advance asynchronously.

For a fixed session and execution contract, write an installed fact as

$$C = (v, F, Z_C, P_C, T_C). \tag{1}$$

Here v is its version, F its certified audio frontier, Z_C the complete selected output, P_C the feedback state for later calls, and T_C its bound display text. A contract specifies the model, adapter, decoding configuration, audio identity, and transformations from a complete output to P and T. Here F is a state-processing boundary. Consumer coverage follows the adapter's publication policy; advancing F need not add visible characters. Certification means that the protected selection and installation protocol accepted this fact. Recognition accuracy against human transcription is a separate property.

The adapter declares a conservative representation of *all mutable cross-call state that can affect later model inputs, selection, or feedback*. Telemetry and device resources are recorded separately. The Qwen adapter passes prompt tokens between calls and carries no numerical KV cache across them. The native correction adapter additionally carries audio buffers, token groups, context, alignment position, and decoder configuration. A state comparison is meaningful only within its declared contract: text equality cannot stand in for an unrecorded alignment cursor or numerical cache. The representation may include more fields than a particular next call needs; we do not require it to be minimal.

B. Three lifetimes of provisional work

A complete provisional node n records a frozen call

$$q_n = (a_n, P_n^-, b_n, \chi_n), \quad n = (q_n, Z_n, P_n^+, T_n), \tag{2}$$

where a_n is the input-state frontier, $b_n > a_n$ the target audio frontier, and χ_n binds the remaining inputs and execution identity. The model may read the complete audio prefix through b_n; a_n does not claim a token-to-audio alignment. P_n^+ and T_n are derived from the complete result through the contract's adapter. An in-flight call has a frozen input and no complete output yet. Partially generated or cancelled tokens cannot become a complete node.

Such work has three distinct lifetimes. Its *dependency lifetime* determines whether it can still be used under the current Canonical root. Its *visible lifetime* determines whether the Consumer keeps its whole published hypothesis. Its *physical lifetime* ends when the executor has actually finished or drained the work. A new root can change the first two without immediately ending the third. RevTrace connects their decisions while keeping the evidence for each decision explicit.

The service retains every accepted input obligation until certified processing covers it. Discarding a branch, requesting cancellation, or recovering an old view does not remove that obligation or reset its arrival clock. In exact mode, a missed deadline never authorizes an uncertified Canonical transition.

C. Trust and failure boundary

We assume one trusted Owner per session, a protected selection path, immutable accepted audio and model configuration, unambiguous encoding, collision-resistant digests, and storage that honors successful synchronization. Calls used in the dependency argument must actually consume their frozen inputs and return the corresponding complete results. This execution-integrity premise is explicit: host receipts and text hashes bind API objects; they are not GPU arithmetic attestation. Private and Authority need not return identical results from independent floating-point executions.

At its declared interface, the runtime checks result identity, completeness, derived-state consistency, and ancestry, including late or repeated messages. Provisional content can still be inaccurate. The separately frozen recovery configuration handles whole-process termination on the same intact filesystem. A compromised coordinator or Authority, Byzantine host code, arbitrary persistent corruption, and lost storage are outside this boundary. The controlled wrong-input injections in Section V intentionally violate normal Private input fidelity; we use them to study containment and executor release, not as examples covered by the lawful-continuation proposition.

III. Valid Continuations and Whole Views

A. A continuation under a new root

The central question is whether an old call still belongs to a valid provisional continuation after trusted feedback advances. Its birth version records dispatch history. The dependency witness establishes whether it remains usable now.

Let $x = (F, P_C)$. A complete recorded node provides an edge

$$(F, P_C) \xrightarrow{n} (b_n, P_n^+) \quad \text{if} \quad a_n = F \ \wedge \ P_n^- = P_C, \qquad (3)$$

under the input, identity, and execution contract of Section II. A finite sequence of such edges is a *valid provisional continuation* of C when each successor consumes the preceding complete output state at its actual frontier. Write $C \rightsquigarrow n$ for a complete path to n. A frozen in-flight successor can belong to the same continuation, but contributes no usable result until completion.

Proposition 1 (Sufficient re-rooting witness)**.** *Suppose an old branch contains a complete boundary node at the new certified frontier F, with output state P_C. If its subsequent recorded calls form an unbroken input–output path to the actual branch tip, those nodes are a valid provisional continuation of C. An in-flight successor may continue when its unchanged frozen input is that tip and its target exceeds F.*

Proof. The boundary node provides (F, P_C), equal to the new root's declared cross-call state. The first successor therefore satisfies Eq. (3). Inductively, each next call consumes the preceding complete output at its recorded frontier, with the remaining inputs bound by the same contract. The path reaches the actual tip. The in-flight successor satisfies the next input edge, without yet supplying an output. Re-rooting changes none of these calls or their frozen inputs. □

This establishes dependency validity. The provisional result remains subject to later selection and revision; re-rooting grants no Canonical write authority. The argument also makes no bitwise-replay assumption for a different floating-point execution. Its witness follows calls that actually occurred.

Two short counterexamples explain the required observations. A branch born under v may reach exactly (F, P_C) when $v{+}1$ is installed; discarding it solely because its version is older loses a valid continuation. Conversely, equal displayed text can hide different feedback tokens or alignment state. A call consuming one state has no valid path from the other merely because their text agrees. Equal state at a different audio frontier likewise does not provide Eq. (3).

B. Retaining computation: R

In the core Qwen implementation, the computation guard implements a conservative witness for Proposition 1. On a Canonical installation, it first finds a complete Private node at exactly F whose full feedback tokens equal P_C. It follows complete successors, checking each frozen input and frontier, until it reaches the recorded branch tip. An intermediate match cannot stand in for a path to that tip. If a successor is in flight, the guard requires the unique tracked task to consume this tip and target audio beyond F.

When the witness succeeds, the Owner retains the branch and updates its eligibility relative to the new root. The original job identifiers, birth versions, input tokens, and results remain unchanged. Otherwise it fences the continuation lacking this witness and rebuilds from an allowed state. This guard's boundary-node and unique-slot requirements restrict what this implementation can retain; they are not universal necessities for every valid reuse scheme.

C. Retaining an actual view: S

Computation retention alone does not preserve service that has already reached the Consumer. A snapshot Consumer can replace an ahead view with the newly certified, shorter view even when the ahead computation is still valid. RevTrace instead maintains a certified lane and a provisional lane. Both hold whole hypotheses; there is no uncertified suffix splice. The Consumer records the certified text and provisional visible

text separately, so a retained provisional view acquires no certified-prefix status.

On applying a new Canonical publication, S checks the complete currently visible provisional node. It can prove a path directly when that node's frozen input is (F, P_C), or follow its complete published ancestors to that boundary. Missing, cyclic, or ambiguous ancestry is rejected. This check uses the same feedback-state contract as R, while accepting a different sufficient witness: a directly rooted published call does not need an extra Private boundary node.

The Consumer first previews the whole next view. The protocol kernel then checks the installed durable root, actual registered results, dependency path, and bound text/frontier before authorizing the actual application. Only that application yields the recorded view receipt. A caller-supplied compatibility flag cannot replace the underlying nodes.

Proposition 2 (Continuity at a compatible application)**.** *Immediately before an authorized Canonical application, let the visible whole provisional view h be ahead of its new frontier F. If S proves $C \rightsquigarrow h$ and no independent withdrawal mode is invoked, that application advances the certified lane and preserves h unchanged.*

Proof. The compatible-and-ahead branch installs C into the certified lane and retains the existing provisional node, text, and frontier. The kernel authorizes this preview against the actual root and complete dependency records before the Consumer applies it. Thus this application does not itself withdraw any part of h. □

An incompatible update must still withdraw the view. Later valid results can also revise text or replace a hypothesis. Consequently the property concerns one compatible root application, rather than global monotonic display. Snapshot-only service and conservative recomputation can be safe; the four-arm experiment asks what additional progress R and S preserve together, and where that benefit is worth its cost.

D. Native feedback and conditional full calls

StreamCorrect's original recognizer, learned correction, and feedback update provide a concrete instance of the contract in Section II [7]. Let P be a complete closed native state and X the next accepted audio input. We expose the recognition boundary as

$$A(P, X) = (B, M), \tag{4}$$

where B contains the ordered beam candidates and M contains the live engine state and inputs required by the remaining correction and update. The protected original C/U selects a correction and produces the next closed state. Its selection, candidate order, prompts, returned token IDs, and update rules remain intact.

Before that selection finishes, the original update tail supplied with a typed first-beam identity prediction produces a provisional closed state $\widehat{P}^{+}$. This projection has explicit provenance: recognition has executed, but learned correction has not completed for this projected node. It supplies a conditional input for the next complete call. Both GPUs host the original full recognizer–corrector pipeline, so a call dispatched on this input executes the protected original C/U as well. Completed conditional full results and typed identity projections remain distinct objects. Neither is installed merely because its predicted text looks plausible.

When the selected parent arrives, the Owner compares the call's entire frozen start state, actual frontier, execution contract, and session epoch with the installed parent. An exact match confirms the already owned call: an in-flight call continues, or an already completed result becomes eligible for ordered installation. The job's birth root, identifier, epoch, inputs, and execution history are unchanged. The call's own identity prediction need not equal its eventual corrected output; confirming its input establishes eligibility, while original correction determines its result.

An earlier ancestor mismatch removes support for the tracked provisional chain, but does not prove that a later complete input cannot converge. Cancellation before correction therefore requires an actual selected state at the call's same start frontier that disagrees with its complete input, coverage of its target without remaining use, or a changed epoch. A still unresolved future parent supplies none of these proofs.

This direct native witness instantiates the continuation relation of Section III. It differs from the core Qwen computation guard, which conservatively requires an explicit matching boundary node and a uniquely tracked successor. These are different sufficient implementations of eligibility. The whole-view Consumer likewise keeps certified and provisional provenance separate and applies complete supported hypotheses. We compare conditional full-call retention with an already-ready ASR packet alternative. Both are author-built implementations of the same feedback principles; the measured native benefit concerns confirmed complete calls.

IV. Execution Protocol

A. Complete results and physical ownership

This section describes the core ASR runtime. The native correction adapter uses its separately declared full-state interface; its validation does not inherit the core kernel's recovery or physical-lease evidence automatically. Figure 2 separates publication from physical completion. Each dispatch binds a session, worker incarnation, job, frozen input, target frontier, and execution contract. The wrapper accepts only an exclusive complete-row or cancelled-row outcome. It re-derives feedback state and text from the bound complete result. A cancelled row reports accounting information and supplies no partial feedback state.

The Owner can validate, durably install, and publish a complete row while other rows of its physical batch are still running. This prevents a completed session from inheriting another row's entire completion delay. Physical ownership has a separate condition: every row must have a terminal outcome and the device must synchronize before `BatchDrained` releases the batch lease. A complete publication alone cannot authorize reusing the executor.

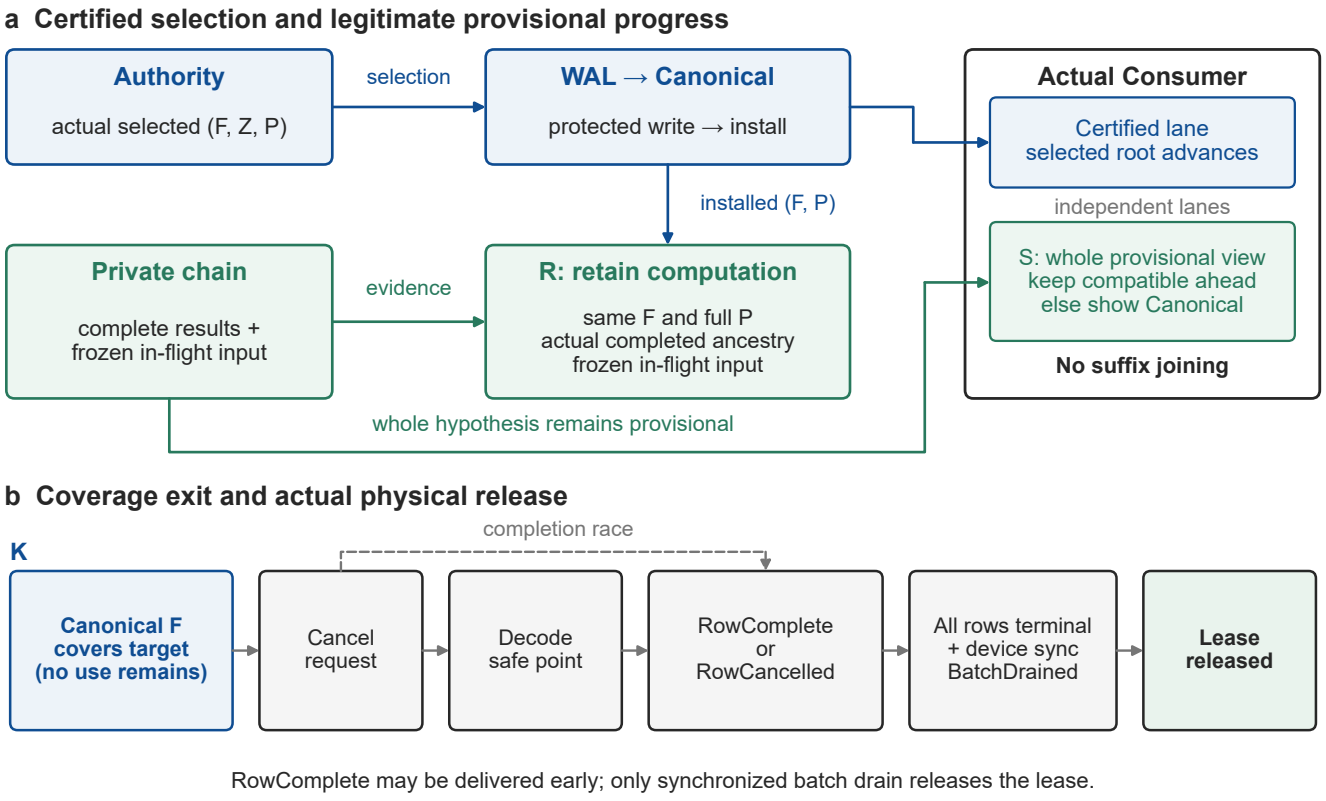


Fig. 2. RevTrace coordinates complete provisional computation and whole Consumer views with certified state. R and S use the same declared feedback dependencies, with separate sufficient guards. Coverage retirement uses a different predicate. A complete result may be consumed before its batch drains; only terminal rows plus device synchronization release the physical lease.

B. Covered work and safe-point exit: K

An old birth version does not determine either usability or uselessness. For the tested use model, K requests exit when the actually installed Canonical frontier covers a proposal's target and that work has no remaining declared use. This coverage predicate answers a different question from R: a valid continuation beyond the new root may still be useful.

`CancelRequest` is a request, not a completion event. The worker observes it at supported decode boundaries. Completion can win the race; the runtime records either a complete result or cancellation, never both. It waits for all row terminals and actual drain before releasing the lease. The static batch is not compacted or refilled, and residual cost for masked rows remains in the measurements. The protocol supplies no arbitrary-GPU millisecond cancellation guarantee. Its service value is the measured execution window available to subsequent lawful work.

C. Durable facts and outstanding input

Canonical state and transition identity are synchronized before a successful acknowledgment. Each journal frame binds the parent and next state, request identity, selected complete result, and receipt. The runtime verifies that the recorded delta reconstructs the proposed state. Group-ready commit shares a synchronization barrier among already validated transitions of different sessions; it introduces no collection timeout and no cross-session all-or-nothing transaction. An uncertain append or state-application outcome blocks further writes until recovery.

Recovery validates framing, checksums, state chaining, and contracts, then reconstructs certified state and admitted identities. An identical retry retrieves its recorded outcome; a changed request with the same identity conflicts. These are conventional write-ahead recovery principles [8] instantiated at the feedback boundary.

In the recovery configuration, the recovered view can be applied before cold neural workers finish loading. This restores existing service facts, not progress on new audio. The input journal separately preserves accepted but uncertified audio and its original clocks, so replaying old text cannot discharge new work. Process epochs fence old workers and publications. These checks rely on exclusive ownership and intact synchronized storage, as declared in Section II.

D. Validation boundaries

The re-rooting argument and conditional view property are supported by targeted source-level witnesses and actual service interleavings. Separately, bounded exploration of an earlier recovery prototype covers durable-root, view-application,

and physical-lease interleavings. Its 1,154-state single-session product does not represent the full multi-node re-rooting guard or an arbitrary neural execution. We therefore use it as component-level protocol evidence, with its own frozen source, rather than a complete machine proof of Propositions 1 and 2.

A separately frozen recovery prototype completed 84 GPU service runs: 24 normal, 36 process-group SIGKILL, 16 controlled-boundary, and eight post-sync crashes. The 44 crash audits recorded no acknowledged-input loss. All 32 early-recovery crash runs reapplied old facts before workers were ready. In one Qwen configuration, median old-view recovery changed from 31.013 to 14.736 s; first new certified output still took 33.829 s. The complete-composition control shared the benefit. All 44 crash runs missed whole-run D2/C8. These are earlier-prototype measurements, not fault validation of the current core or native adapter.

V. Evaluation

A. Experimental design and service endpoints

We evaluate two service obligations under unchanged selected feedback histories: retaining dependency-valid progress already at the Consumer, and completing selected transitions before their original deadlines. The core factorial study isolates computation and whole-view retention; the native study tests input-confirmed full execution of the original correction pipeline. Timing, content support, and energy remain separate. The core experiments use actual neural calls, durable input and Canonical journals, and an instrumented FIFO Consumer. Measurements use Consumer applications rather than publication or display replay. Accepted input obligations retain their original arrival clocks.

a) Implementation and workloads: The service uses two 24-GB RTX 3090s: an FP16 Private worker and an FP32 Authority worker. Qwen3-ASR 1.7B [9] is the primary backend; Whisper base.en [10] tests a second adapter. The common carrier uses complete-row delivery, at most 16 rows per batch, a 256-token generation limit, the protocol kernel, and synchronized journals. These experiments use `row_complete`: physical early exit is disabled, so the effects of R and S cannot be attributed to cancellation. Model startup is outside the serving interval and is recorded separately. Serving energy integrates both physical boards; CPU energy is outside that measurement.

The independent confirmation comprises 24 previously unused speaker streams from LibriSpeech [11], totaling 714.11 s, selected using metadata before inference and partitioned into three eight-speaker cohorts. Its 110 cells contain 60 primary cells (three cohorts, two backends, two arrival patterns, five arms), 20 fresh-process repeat cells on cohort A, and 30 coarser-feedback controls. Primary Authority advances at every 2-s input boundary ($\rho = 1$); $\rho = 3$ changes both selection frequency and feedback segmentation. Synchronized streams start together; staggered streams start 0.25 s apart. Repeats and arms reuse these sources: the independent acoustic sample is 24 speakers, not 110 deployments.

b) Comparators: The four arms cross computation retention R with whole-view retention S (Section III). With R disabled, computation rebuilds from the selected root rather than retaining its witnessed continuation. With S disabled, a Canonical application replaces the provisional view with the new certified snapshot. This *checked-snapshot* Consumer still validates Private dependencies against the same root and passes the same protocol kernel. Removing retention does not remove validation, durability, input accounting, or the execution capability of Section IV. The fifth arm, *Full composition (ours)*, is a complete-composition positive control: an alternative policy instantiation with the same retention relation, Consumer protection, and shared execution/commit substrate. It is neither an external pre-existing implementation of our complete design nor an independently reimplemented neural stack. Comparable behavior is the expected positive-control outcome.

c) Endpoints: For offered interval i, let w_i be its audio duration, r_i its original ready time, d_i its first actual visible covering time, and c_i its first actual certified covering time. Original joint-deadline credit is

$$A_{2,8} = \sum_i w_i \, \mathbf{1}\{d_i - r_i \le 2 \ \wedge \ c_i - r_i \le 8\}. \qquad (5)$$

All offered audio, including end-of-stream (EOS) tails and misses, stays in its denominator. Whole-run service passes only when display P99 is at most 2 s and certified-view P99 at most 8 s. P99 is computed over offered frontiers within each run; credit weights their audio durations. Thus a P99 pass can coexist with a small tail miss. We separately measure adjacent visible-frontier withdrawals, their audio magnitude, and the integral of the deficit below the previously observed high watermark. Text revisions, final word error rate (WER), and board energy remain separate endpoints. Independent recounters check complete selected $F/Z/P/\mathrm{text}$ histories and reconstruct dependency witnesses from actual complete outputs and consumed inputs, rather than trusting a retention flag.

B. Independent confirmation of the composition

Table I reports the pre-specified Qwen, $\rho = 1$, staggered stratum, including all three cohorts. Retaining computation alone raises deadline credit from 654.115 to 664.150 s but increases audio-frontier withdrawals from 12 to 17. The additional work survives the new root, while snapshot consumption still discards its visible progress. Retaining views alone reduces withdrawals to one but yields 628.435 s of credit. The joint arm obtains 680.760 s and zero audio-frontier withdrawals: 26.645 s, or 3.73 percentage points of offered audio, above $R_0 S_0$. Its gain over computation-only is 16.610 s.

The cohort-wise credit interaction, $I = Y_{11} - Y_{10} - Y_{01} + Y_{00}$, is $+18.530$, $+6.205$, and $+17.555$ s. These three contrasts share queues within each cohort. They support a composition effect in this stratum, rather than treating individual deadline windows as independent trials. The joint arm still erases 569 visible characters: zero audio-frontier withdrawal does not mean immutable text. Its 2.329-s median display P99 also remains above D2. The result preserves more timely progress without

TABLE I
INDEPENDENT QWEN CONFIRMATION, $\rho = 1$, STAGGERED ARRIVALS. EACH ARM OFFERS 714.11 S ACROSS THREE EIGHT-SPEAKER COHORTS. WITHDRAWALS AND CREDIT ARE TOTALS; P99 AND ENERGY ARE MEDIANS OF THREE PER-RUN STATISTICS. ALL 15 RUNS MISS THE WHOLE-RUN D2 CRITERION.

Arm	Audio withdrawals	Display P99 (s)	D2/C8 credit (s)	Energy (J/audio-s)
R_0S_0	12	3.797	654.115	59.72
R_0S_1	1	3.891	628.435	59.16
R_1S_0	17	2.431	664.150	59.68
R_1S_1	0	2.329	680.760	59.37
Full (ours)	0	2.534	667.530	59.73

establishing that this workload meets its whole-run service target.

The full composition likewise has zero frontier withdrawals, the same 569 erased characters, and comparable timing and energy. Its lower credit than the joint arm in this stratum is not evidence that a different complete composition is inherently inferior. Across the confirmation, all 144 matched source/backend/granularity/arrival history groups agree on complete Canonical $F/Z/P/\mathrm{text}$; all 22 paired full-composition comparisons preserve those histories. The ablations therefore isolate how legal progress is retained, rather than buying latency by changing the selected feedback task.

C. Relative progress explains when retention matters

The useful opportunity is an already observed whole view ahead of a new Canonical frontier whose complete dependency path remains valid. In the headline joint arm, 18 of 368 Canonical updates encounter such a view and retain it. Computation-only encounters 17 and clears all 17. Under synchronized arrivals the joint arm encounters only three of 368; its credit is 0.790 s below computation-only. Without an ahead view, preserving it has little service value. These are witnessed incidences in the executed workload, not a production prevalence estimate.

Timing can also change the ranking. In the fresh-process staggered repeat on cohort A, joint credit is 226.320 s versus 230.565 s for computation-only, and $I = -2.245$ s. We retain this repeat alongside the three positive primary contrasts. The combination's value depends on relative progress and actual feedback compatibility; independent source confirmation does not imply an invariant timing advantage.

D. Operating workloads and subsequent timely continuity

We next freeze three development workloads—eight synchronized sessions, eight staggered sessions, and sixteen staggered sessions—with four arms and three fresh processes per arm (36 cells). A single later exploratory midpoint adds twelve staggered sessions and the same twelve arm/repeat cells. Figure 3 retains all 48 runs. The nested source lists offer 121.205, 185.300, and 250.180 s per eight-, twelve-, and sixteen-session run. Adding sessions also adds source content; these are explicit operating workloads, not a pure concurrency capacity curve.

All arms pass whole-run D2/C8 in the two eight-session workloads. Joint display P99 is 1.426–1.528 s when synchronized, where the arms largely converge. Staggered joint P99 reaches 1.955 s, and an earlier development run reached 2.034 s, exposing a narrow timing margin. Two 0.215-s tails exceed D2 in the new eight-session runs and remain uncredited even though their runs' P99 passes. At sixteen sessions, all twelve runs fail D2, while C8 passes. Joint credit exceeds R_0S_0 by 20.280/22.885/20.180 s, but its credit interactions are $+20.910$, $+26.380$, and -9.575 s. Against computation-only, joint credit changes by $+18.910$, $+5.055$, and -7.225 s, and display P99 improves in only one repeat. Continuity is the stable distinction: computation-only withdraws compatible views 16/17/12 times, versus zero for joint.

First credit does not reveal a later withdrawal. A frozen secondary recount therefore follows intervals already delivered within D2 whose eventual certified covering view also meets C8. At each subsequent Canonical application it classifies withdrawal using the complete old view's dependency witness, separately recording necessary incompatible withdrawals. Loss lasts until an actual covering view returns or the next Canonical boundary censors this root's window. At sixteen sessions, computation-only still has 10/12/9 timely compatible loss windows with 24.903/23.635/15.736 audio-s×wall-s deficit area; joint has none. Such holes intersect the original D2 read point for 6.000/10.675/4.000 s of distinct audio. The secondary measures continuity after delivery; it neither changes Eq. (5) nor certifies provisional content.

The twelve-session midpoint sought a stronger witness: both computation-only and joint must pass whole-run D2/C8 in all three repeats, computation-only must lose an originally timely complete 2-s non-EOS interval in at least two, and joint must have no compatible timely loss. This witness is *not established*. All twelve midpoint runs fail D2; joint P99 is 2.197/2.299/2.228 s. Computation-only has 1/1/3 timely loss windows, but none qualifies as a complete non-EOS interval. Joint has zero, yet cannot turn this into a successful whole-run operating point. The predeclared stop rule closes this exploration without another session count or extra repeats.

E. Quality and cost boundaries

All headline Qwen arms have WER $26/1908 = 1.36\%$ and identical selected histories. Retrospective support counting first-certified content plus first-ahead whole views that exactly match later same-F Canonical text leaves the headline credits unchanged. Elsewhere raw differences remain: joint loses 2 s of strict support in the sixteen-session third repeat and 2 s in each of the midpoint's first two repeats, although the frozen text normalizer agrees. This comparison is with the selected model output, not human-transcription correctness. Whisper exposes a larger adapter boundary: WER is 83.12% at $\rho = 1$ and 11.53% at $\rho = 3$, whose different feedback contract prevents interpreting it as a quality-preserving speedup.

The headline arms consume similar board energy (Table I); energy does not rank them in credit order. At sixteen development sessions, joint reduces total model time by 0.66–1.59 s

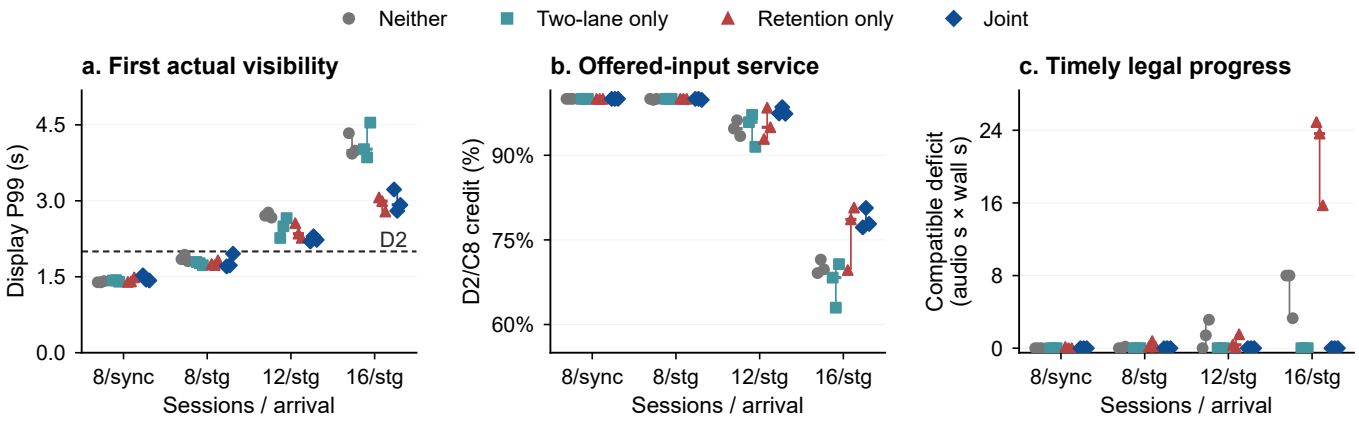


Fig. 3. All four development workloads and three fresh-process runs per arm, including the negative twelve-session midpoint. Points are real per-run values; ticks and ranges describe their median and observed range. Panel (c) is the separate secondary measure of subsequent compatible loss of originally timely intervals, censored at the next Canonical root when necessary. It does not replace panel (b)'s original first-observation credit.

against R_0S_0, while serving wall time does not consistently fall and the three energy differences change sign. The measured benefit is the progress delivered and preserved under a fixed feedback contract. It is not a general reduction in inference cost or a cure for overload.

F. Native correction with the original selected task

We retain StreamCorrect's published recognizer, learned correction, and feedback update [7], using the fixed 500-ms, no-VAD adaptation and original FP32 execution. Both compared policies have two protected full workers, the same ASR early-view Consumer, and identical neural warmup. Conservative execution waits for selected parent feedback before starting the next call; full-call retention confirms conditional calls as in Section III-D.

a) Development and frozen confirmation: Component controls cover all 69 steps of four AISHELL-1 utterances used for implementation development [12]: 65 complete ASR packets reproduce original C/U, with four buffer-only steps using the original path. Thirteen identity-projected ends differ from selection without invalidating that candidate input. Comparison with the earlier implementation removes only the newly added implementation-contract tag; complete task state, corrections, and returned IDs remain equal.

A three-arm source-0 pilot gives conservative execution, already-ready ASR handoff, and conditional full calls 3.0, 6.5, and 8.0 s of original credit over 8.068 s offered. Full calls can retain work still executing when its input is confirmed; ready-only handoff requires a complete received packet. We select our full-call implementation and confirm it with 16 runs on all four development sources. These are development measurements, including two 0.068-s EOS misses, rather than independent acoustic evidence.

We then freeze that method before selecting four previously unused AISHELL-1 speakers using archive metadata and an 8–9-s duration band. All four initial selections remain; no model output selects a source. The final experiment has four separate chains, two fresh-process repeats, and two policies: 16 runs. Paired arm order reverses on the second repeat, GPU roles swap, and pair blocks are randomized. The independent acoustic unit is a speaker, not a repeat or offered frontier. All original 500-ms frontiers and partial EOS intervals stay in the denominator.

TABLE II
NATIVE SERVICE: EIGHT RUNS PER ARM IN EACH PANEL. OFFERED AUDIO PER ARM IS 66.430 S FOR DEVELOPMENT AND 66.340 S FOR INDEPENDENT SOURCES. CREDIT AND ENERGY ARE TOTALS; C50 IS THE MEDIAN OF PER-RUN C50 VALUES. TEXT SUPPORT COMPARES THE FIRST ACTUAL WHOLE VIEW AT G WITH LATER COMPLETE CANONICAL TEXT AT EXACTLY G. VALUES ARE ROUNDED.

Policy	Credit (s)	Text support (s)	C50 (s)	Energy (kJ)	Pass
Development: four existing sources					
Conservative	37.500	33.500	1.905	36.942	0/8
Full-call	66.294	56.288	1.035	32.050	6/8
Independent: four new speakers					
Conservative	37.000	29.000	2.115	35.801	0/8
Full-call	66.340	51.518	1.203	31.815	8/8

b) Task fidelity and timely service: Offline original full calls supply 68 reference transitions, separately from service timing. All 272 Canonical transitions in the new experiment reproduce their complete original start and end states, C, and returned IDs, including the implementation tag. Both arms have five character edits over 81 unique reference characters, or 6.17% CER; across repeated flows the equivalent count is 10/162. This measures the adaptation, not the published author's benchmark CER. Development runs separately preserve all 276 transitions and have 16/182 repeated-character edits.

Table II keeps development and independent results separate. On new speakers, credit increases in all eight pairs, from 37.0 to all 66.33975 s offered per arm. All four speakers have a positive two-repeat sum, satisfying the frozen directional rule;

this is not an eight-independent-sample significance test. Full-call retention passes whole-run D2/C8 in 8/8 runs, versus 0/8. Its D99 ranges from 0.837 to 1.663 s and C99 from 1.090 to 1.929 s. It has no D2 or C8 miss; conservative execution has 62 D2 misses, including eight EOS intervals, and no C8 miss. The median per-run C50 falls from 2.115 to 1.203 s. Both-board serving energy, excluding loading and fixed neural warmup, falls 11.13%; credit, C50, and energy improve in every pair.

c) Retained work and visible content: Ninety-two promoted calls are actually computing before their selected parent finishes, and their model intervals overlap the parent's. Twenty have their own identity-predicted end differ from original correction. Twenty-four calls stop after ASR under invalidity or coverage proofs, before C/U; two already-completed conditional full results are discarded. None changes the original selected history. These observations distinguish input confirmation from requiring each future prediction to be correct.

The new runs also contain a concrete whole-view witness: one Canonical application advances from 4.0 to 4.5 s while retaining a supported 5.0-s Private view and its original executing call. Two other applications, on one speaker's two repeats, withdraw 6.0 to 5.5 s because the call's complete predicted input disagrees with the actual selected state; the two complete conditional results are not installed. These natural events illustrate compatible retention and necessary withdrawal, not another factorial estimate of the $R \times S$ interaction.

Posterior whole-text support remains separate from original credit. Supported credited audio increases from 29.0 to 51.518125 s in all eight pairs; raw and whitespace-only comparisons agree and unknown coverage is zero. The remaining 14.821625 s of full-call credit does not have this support. Six of eight first EOS texts differ from later Canonical text in each arm. These are selected-model comparisons, not human correctness certificates. Both arms introduce 180 LCP-versioned codepoint occurrences, retract 14, and end with 166 across eight flows. Unchanged-text rows first cover 10.0 versus 17.0 credited audio seconds; processing-frontier credit does not count newly transcribed characters.

An earlier two-source capacity experiment retains its negative result: retention earns 17.0 s versus 18.5 s for independent full replicas, using 47.991 versus 25.949 kJ. Its warmup and Consumer differ. The present single-chain comparison establishes a service opportunity in this scope; it neither supplies a causal before/after comparison nor establishes superior saturated multi-source capacity.

G. From physical release to supported service

We isolate computation retention R and coverage exit K while enabling the legal two-lane Consumer S in every arm. The study uses the same executor, 16 fixed development sources, and 250.180 s of offered audio per run. Four arms, synchronized or staggered arrivals, and three fresh-process repetitions give 24 runs. A frozen wrong-conditioning fixture supplies controlled useless tails. It is separate from the unmodified-input work used to establish lawful continuation. Original input obligations, D2/C8 deadlines, quality, and physical costs remain unchanged.

All selected Canonical $F/Z/P/T$ histories match, with WER $14/687$ in every run. All 675 pairs with identical complete processed inputs, actual prompt/audio identities, and peer order return identical outputs.

Each synchronized $K1$ run releases a following batch of 16 rows that matches its $K0$ counterpart on these inputs and outputs. Under $R1$, the following model computation starts 473/577/498 ms earlier across the three repetitions. Request, acknowledgment, and drain remain distinct: in one repeat, request-to-ACK is 22.196 ms and ACK-to-actual-drain is 13.359 ms. All 96 following rows across the six synchronized comparisons are actually consumed and first advance their target frontiers.

Canonical may already cover an obsolete proposal, so obsolescence alone does not establish a missed deadline. Comparing actual first coverage from either lane, only 6/6/1 rows under $R1$ and 4/2/6 under $R0$ change from dual-deadline miss to hit. All 96 first-consumed hypotheses agree byte-for-byte with later complete Canonical text at exactly the same frontier, and their worker records contain no input override. This retrospective selected-model agreement supplies content support for the newly timely rows; it is not an online content certificate or human transcription accuracy.

The whole-run boundary is less uniform. Adding K to $R1$ increases original credit by 15.255/13.265/7.970 audio seconds under synchronized arrivals, but changes it by $-26.000/-6.475/+21.780$ s under staggered arrivals. Neither whole-run model time nor board energy improves consistently. Whole-run first content remains revisable: each synchronized arm has 16 s of timely first-ahead content differing from later same-frontier Canonical text, or 14 s after the fixed normalizer. These equal deductions leave the synchronized credit differences intact. We therefore use simple coverage exit as execution support, with a measured drain-to-service chain, rather than a universally beneficial cancellation algorithm or a stable positive $R \times K$ interaction.

VI. Related Work

a) Incremental information and dependency reuse: The incremental-unit model separates newest committed information from the active right edge and propagates dependency revocation [1]. Its commit promises that a unit will not be revoked. Here, certification records acceptance by the protected selection protocol; later selected facts may still revise the transcript. Adapton checks recorded dependencies; Nominal Adapton combines names with arguments and dependency checks to recover reusable work [2], [3]. Differential Dataflow maintains incremental computations across evolving versions [4]. These works establish much of the basis for dependency-aware reuse. RevTrace instantiates that basis for complete neural feedback state and frozen executed calls. Its service obligation connects execution eligibility to the actual whole view retained when an independently selected root is applied. Our sufficient witness follows recorded executions; it does

not transfer a from-scratch theorem to independently repeated floating-point inference.

b) Revisable speech services and speculation: Whisper-Streaming converts an offline recognizer into incremental service using LocalAgreement [13]. StreamCorrect adds learned correction and feeds selected corrections into subsequent recognition [7]. RevTrace retains that selection and feedback behavior in its controlled native adapter; its contribution concerns the asynchronous execution and consumption around the feedback boundary. It does not introduce a better acoustic model or train a new corrector.

SSBD coordinates old-output draft reuse with display-only suffix masking [5]. It revalidates a draft under updated input and includes an unbiased configuration. RevTrace follows complete calls that actually executed to establish their eligibility under newly selected feedback state, then checks the dependency witness for the Consumer's whole current view. PredGen provides a close precedent for speculation while speech input evolves across recognition, language generation, and speech synthesis [6]. Our comparison holds the complete selected feedback task fixed across the retention policies. The specific reuse object, protected selection, and whole-view transition define the composition evaluated here. RelayS2S uses a learned verifier to commit a fast-path prefix that constrains subsequent slow-path generation [14]. Our native calls instead confirm their complete inputs and retain the original correction and feedback update. Work on ASR alignment drift also establishes the importance and cost of audio position state [15]; our contract includes that state rather than treating text equality as sufficient.

c) Execution, authorization, and recovery: Retire studies versioned execution and bounded physical exit for interruptible agents [16]. RevTrace likewise distinguishes logical progress from an executor's actual release, while deriving the tested retention and coverage decisions from the feedback contract. Our simple covered-exit implementation is execution support, without a claim to a new general cancellation algorithm. A parent-version cancellation control is not a reproduction of the complete Retire system.

Optimistic and multiversion concurrency control separate validation and commit from concurrent execution [17], [18]. Write-ahead recovery persists state changes and identities [8], and stream-processing systems coordinate recoverable state [19]. We use these established principles at the ASR feedback boundary. They supply machinery for ordering and durability; the adapter still has to specify which state constitutes a valid continuation and which complete view the Consumer may retain.

VII. Discussion

a) Scope of the guarantee: The continuation witness assumes a sufficient adapter state and faithful execution of frozen inputs. A new adapter must account for every mutable field that can affect later inference, selection, or feedback. A hidden KV cache, alignment cursor, or external tool state can invalidate a token-only argument. Conservative rejection remains allowed when a sufficient witness is absent. The guarantee concerns dependency validity and an individual compatible application; it supplies neither transcription correctness nor monotone text for an entire session.

b) Progress, content, and capacity: Processing more audio can leave the cumulative transcript unchanged. Frontier credit, actual text additions and revisions, eventual selected text agreement, and ground-truth accuracy answer different questions. We report them with their original denominators. Preserving compatible progress also cannot remove all input synchronization, queueing, or neural work. The negative c12 witness and the native two-replica comparison identify concrete limits of this implementation rather than a universal capacity envelope. The independent native confirmation establishes a single-chain service opportunity with the original selected task; it does not turn that result into saturated multi-source capacity.

c) Deployment and validation: The prototype evaluates short streams selected before inference and does not establish bounded memory or long-running collection of dependency history. Production deployment would also require network-facing delivery and ownership transfer beyond the local FIFO Consumer. Our crash evidence uses an earlier frozen recovery prototype on intact storage; it is not a full-stack proof for every later runtime or native adapter. The artifact separates source-level controls, bounded protocol exploration, controlled faults, actual GPU service, and retrospective analyses so their evidence roles remain inspectable.

VIII. Conclusion

RevTrace preserves dependency-valid speculative work and whole delivered hypotheses while trusted feedback continues to advance. Its explicit feedback contract supports executable re-rooting witnesses, coordinated computation and consumption, and separate physical retirement. Actual service experiments show that first delivery can meet its deadline while subsequent compatible updates withdraw the delivered progress. The joint design preserves the whole ahead view at compatible root applications, with necessary revisions, execution costs, and capacity limits kept visible. The core experiments show why retaining computation does not itself preserve already delivered service. The native implementation confirms a call's complete input and retains original correction work without requiring its own next prediction to be correct. Together, these results establish a concrete composition of dependency reuse, protected feedback, and whole-view consumption, with measured timing and capacity boundaries.

AI Tool Usage

ChatGPT/Codex assisted literature discovery, experiment orchestration, analysis and validation code, manuscript drafting, and figures. The artifact records execution sources, frozen configurations, controls, and result provenance, and distinguishes actual service measurements from component tests and retrospective analyses. The authors retain responsibility for the submission.

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